Research statement

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Abstract

Erratic rainfall due to climate change along with depleted groundwater tables poses a significant threat to the sustainability of agriculture in India, on which nearly half of the rural population depends for its livelihood and sustenance. In response, various welfare and social protection schemes in India fund the construction of water assets such as farm ponds and check dams, through participatory planning processes with rural communities embedded within local governance institutions. However, persistent challenges remain in such participatory planning processes [1], such as inequities in participation, instances of spurious and non-functional assets, and top-down guidelines that hinder context-sensitive planning. To address such challenges, I led the development of the underlying geospatial dataset infrastructure and the overall design of *Commons Connect* [2], a digital participatory tool designed to improve transparency and build community consensus to improve equity in asset allocation. The tool provides various geospatial datasets generated at village-level planning scales to identify viable and useful locations for assets [3]. Building on this, I developed an impact projection framework [4] as a digital-twin to simulate climate resilience under different what-if scenarios of planning. Finally, I built a scalable system for knowledge representation and transfer [5] that enables the identification of localized landscape stresses using dozens of social-ecological indicators across diverse production systems and regions in India. Together, these contributions aim to bring evidence-based, equitable and climate-resilient community-led planning for rural areas. Tools like *Commons Connect* founded on my research are now being widely adopted and used across nine states in India. The following sections explain each of these contributions in detail, along with efforts in other projects mentioned in section 5.

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1. Improving participation in planning

Inequitable allocation of water assets remains a persistent challenge in participatory planning processes. Through field visits in the Bihar and Jharkhand states of India, I conducted interviews and focus group discussions with multiple stakeholders through which we observed that demands for water assets from marginalized groups are often under-represented [1]. This is difficult to address due to elite capture at local governance levels where socially and economically dominant groups influence the articulation and allocation of demands. To address this, I led the development of *Commons Connect*[2], a digital participatory tool that enables field staff and volunteers working with Civil Society Organizations (CSOs) to digitally record demands of water assets, including those owned or used by marginalized communities, and support them using relevant geospatial datasets to justify the need for these assets. The demands and evidence of effectiveness of the proposed assets are included in standard documents called Detailed Project Reports (DPRs) which are submitted to government schemes for fund allocation. As a result, these digital trails of community demands help make historical inequities more visible and legible. This work also engages with a broader question of whether technology can strengthen social cohesion and foster collective action by improving transparency and accountability in planning processes. Commons Connect has supported the development of 500+ DPRs so far (June 2026), with an additional 1,000+ in progress across villages in 9 states of India. Among the developed DPRs, 60% demonstrate improved equity in planning, along with a shift toward data-backed, community-driven demand for water assets.

2. Enabling data-driven planning

The creation of non-functional or ineffective assets that fail to support adequate storage or groundwater recharge remains a critical challenge in field-level planning. This challenge stems from fundamental uncertainty in identifying the most relevant biophysical variables that are necessary for effective planning, compounded by the absence of integrated and fine-scale data at the planning unit. In practice, significant time and effort by CSO field staff is spent on ad hoc curation and compilation of fragmented datasets, limiting the consistency and scalability of planning processes. To address this, we developed a host of datasets [3] using remote sensing, machine learning, and hydrological modelling. These datasets span multiple biophysical domains including hydrology (rainfall, runoff, evapotranspiration, water balance), climatic (drought incidence), terrain (geomorphology, micro-watershed delineation and connectivity, drainage lines), land use and land cover (including intra-annual surface water presence and cropping patters), and tree cover (canopy density, tree height, and other traits), along with welfare and administrative data that includes information on existing MGNREGA [1] assets. Specifically, I led the development of hydrological datasets starting from the delineation of hydrological boundaries across India such as microwatersheds to computing hydrological variables on these boundaries such as runoff, downscaled evapotranspiration, changes in groundwater levels and catchment area, along with climatic and terrain datasets such as drought incidence, monsoon onset, elevation-based landforms. The datasets were generated at the highest available spatial and temporal scales, with landscape-level datasets produced at the microwatershed scale, roughly corresponding to the size of a village in India and site-level datasets at 30 m spatial resolution. Integrated into *Commons Connect*, landscape-level datasets support community consultations conducted by CSO field volunteers to build a data-driven and shared ecological understanding of the landscape, while site-level datasets support site-selection for new assets by providing insights into the site-suitability. Such integration of scientific data into community-led planning is intended to improve the effectiveness of water assets constructed for irrigation and groundwater recharge, and strengthen participatory processes to support equitable and evidence-based planning.

I also participated in the development of a methodology for downscaling evapotranspiration data which can help estimate field-scale crop water consumption to detect water stress and quantify irrigation requirements [6]. Evapotranspiration (ET) serves a proxy for crop water consumption but existing remote sensing data products largely have a coarse spatial resolution. We built a machine learning model to downscale MODIS ET using satellite data from Landsat, meteorological variables, and several land surface

[1] Mahatma Gandhi Rural Employment Guarantee Act that funds soil and water conservation, and drought and flood protection works, among other livelihood and infrastructure assets.

characteristics. My specific contribution in this work was to guide the choice of satellite data products, validate the downscaled data against measured in situ data (lysimeter data from IMD), and build field-level indicators of water stress and water use efficiency.

3. Projected impact of planning

A key challenge in water security planning is to determine how different combinations of water supply and demand interventions can provide reliable and resilient irrigation under increased climate variability and groundwater stress scenarios. Existing planning approaches do not adequately assess whether proposed water structures (supply-side interventions) can reliably support irrigation requirements of ongoing and proposed cropping patterns (demand-side interventions) during events like extended dry spells. This limitation arises because current planning approaches are largely static and non-predictive, such as thumbrules to implement ridge-to-valley approaches or Multi-Criteria Decision Analysis (MCDA) for site assessment. As a result, they cannot precisely answer questions such as (i) What should be the scale of supply-side and demand-side interventions to be planned to ensure reliable irrigation while maintaining sustainable groundwater use? and (ii) With a given scenario of water structures and demand-side patterns, how many irrigations can be supported through surface storage and groundwater pumping during years of consecutive droughts? Addressing these questions requires predictive modelling that explicitly represents hydrological processes and their interactions with climate and farmer responses. My first contribution uses process simulation approaches to project irrigation and groundwater use that can be supported under different planning scenarios, while the second uses machine learning in a social-hydrological setup to model farmer response to interventions, and overall resilience achieved under this response for various climate scenarios.

First, I developed an impact projection framework based on rainfall-runoff simulations to support operational irrigation planning, addressing questions of where interventions should be placed and what scale of intervention is required to ensure reliable irrigation while maintaining sustainable groundwater use [4]. The framework projects the storage and recharge volume of planned water structures starting from a site-level simulation, eventually integrated into a landscape-level (microwatershed) water balance simulation under different planning scenarios. At the site level, this framework helps translate complex equations and numbers into easily relatable metrics like the number of irrigation cycles that can be supported with the structure, additional cropping area brought under drought protection, and which downstream wells are likely to receive groundwater recharge. At the landscape level, it enables the simulation of what-if scenarios that combine different configurations of water structures and cropping patterns to evaluate the overall impact on the micro-watershed water balance. Overall, this framework embeds hydrological intelligence into participatory planning processes, facilitating communities to make informed, data-driven decisions to improve irrigation reliability while supporting sustainable groundwater use.

Second, I participated in the development of a machine learning based decision-support framework to specifically model farmer response to new interventions [7]. It is well documented that new water structures created to improve drought resilience can be misused in the sense that farmers may begin to utilize the additional water to increase their cropping intensity or diversify to higher water consuming crops, rendering the overall landscape again with lower resilience to cope with droughts. In this framework, we computationalized landscapes as social-hydrological systems of dozens of time-series variables, linked together in a dependency graph based on hydrological processes, with downstream variables predicted using machine learning methods applied on upstream variables. The framework models climatic, hydrological, agricultural, and static landscape variables, all derived primarily from remote sensing datasets, and simulates groundwater storage under different climatic and intervention scenarios. My contribution in this work was to model the social-hydrological system as a system dynamics model and design its corresponding machine learning pipelines with other team members, numerically model groundwater fluxes, and generate grounded planning scenarios. The framework can facilitate policymakers to evaluate the long-term impacts of planning on irrigation and groundwater sustainability, thereby improve water resilience under different climate scenarios.

4. Diagnosing landscape stresses for planning

Significant time and effort is spent by CSOs in diagnosing resource-related stresses in a landscape. Diagnosis relies either on field data collection where comprehensive measurements or community discussions conducted by experienced practitioners can bring to light specific gaps in a particular landscape, or requires scientists and researchers who can use secondary datasets to figure out appropriate landscape capacity thresholds and if these are being breached. Both methods require experts and intensive work either on the ground or in modelling. To address this challenge so that problem diagnosis becomes more easily accessible to anyone, I led the development of two complementary data-driven frameworks for natural resource management. The first focuses on characterizing community-reported stresses across rural production systems and describing them using various landscape-level datasets. The second focuses specifically on irrigation-related stress in agriculture and uses remote sensing data to infer the irrigation pattern in an area.

For the first method, we recognized that CSOs through their deep field immersion are able to write diagnosis reports of different stresses in a particular landscape like a village or micro-watershed, but this knowledge sitting in reports is not expressed in any standard format. This lack of structured knowledge representation implies that it cannot be transferred or reused elsewhere either [5]. We leveraged LLMs to extract structured information from these case-studies, such as the location, specific problem diagnosis from a curated dictionary of problem types, and similarly specific interventions and traditional knowledge used to address these problems. We then developed a mechanism to classify these problem types using secondary data known about the landscape, like its stage of groundwater extraction, seasonal surface water availability, deforestation and tree degradation, land-use changes, and other datapoints [5]. By building classifiers for each problem type in a production system using standard datapoints we had effectively built a method of discovering how to represent different problems and their landscape context in data, and use it to also discover other landscapes that would be facing similar issues. While currently we built the classifiers by hand by imposing specific thresholds on relevant landscape indicators, we are now also trying to build a knowledge graph that is extracted automatically by LLMs by reading research papers related to different production systems. We feel that by connecting indicators related to underlying drivers and landscape responses to these drivers we will be able to further enable reasoning on this knowledge graph to trace the problematic mechanisms operating in a landscape. This work can also contribute towards a living knowledge-creation system for natural resource management where continuously evolving ground-level insights are transformed into structured data to support scalable cross-regional learning, data-informed planning and policy design.

Second, we focused on irrigation-related stress contributing to agricultural water scarcity. We developed a decision-support framework for CSO volunteers and communities while planning water assets to improve overall agricultural income of the landscape while ensuring equitable allocation of assets [8]. Agricultural income of a landscape can be proxied through cropping intensity and remote sensed vegetation indexes of crop health, and which in turn are influenced by the availability of irrigation sources such as ponds and wells. The availability and distribution of these irrigation sources determine the resilience and disparities in cropping outcomes in different parts of a landscape. We used computer vision methods to identify wells and farm ponds in an area, and spectral satellite data to estimate cropping intensity and crop health especially during drought years. My contribution was to use this output to model irrigation sources as treatment variables and estimate their causal effect on cropping intensity using causal machine learning. This facilitates the simulation of various what-if scenarios to propose new water assets such that they maximize aggregate cropping intensity while also ensuring equity in productivity with different spatial placements of the irrigation sources.

5. Machine learning on satellite data

I have also worked on developing machine learning models on remote sensing data for various site identification and assessment tasks. I built a method to identify the spatial extent of mining sites by using a deep learning model (UNet) for semantic segmentation in areas around known mining point coordinates of surface mines [9]. We were able to achieve this using open satellite imagery from Sentinel and Landsat sources. I performed various ablations with data augmentations, feature additions, architectural variations, loss functions, and optimizer variations, and eventually provided an extensive social-ecological analysis of mining areas in terms of deforestation, air pollution, socio-economic development, road access, and other indicators of areas around the minng sites, and compared them with similar non-mining areas to determine if mining led to improved development indicators for the immediate affected population living around the mining sites.

I also contributed to building high-resolution land-use and land-cover classification maps for India [10]. We trained pixel-based classification models to classify each pixel into one of the four land-use classes: greenery, water bodies, barren land, and built-up area. My contribution was on data curation and data cleaning methods to build a labelled dataset for India. Initial labels were obtained through the crowd-sourced mapping platform Open Street Maps (OSM) and were sampled across different geographic regions in the country to ensure spatial diversity.

6. Future work

My research experience lies in participatory approaches to technology development, demonstrated through my work on *Commons Connect*, where I co-designed a data-driven tool with various stakeholders to support transparent and inclusive water security planning processes. Building on this, I bring strong domain knowledge in hydrology and technical expertise in machine learning, remote sensing, and geospatial algorithms to develop large-scale datasets and computational planning frameworks for social-hydrological systems. My work is grounded in an interdisciplinary understanding of hydrology and machine learning, which I was able to apply to water security planning, particularly in the context of protective irrigation and groundwater sustainability.

Going forward, I aim to continue addressing research gaps I have come across during my PhD journey in the domain of water security, such as for aquifer mapping and diagnosis of water-related stresses across landscapes. By working in the domain of water security, I built my niche expertise in solving interdisciplinary problems that require a strong domain understanding, which is translated into computational and data-driven models, and packaged in practical and deployable digital tools for real-world decision-making. Beyond water security, I aim to extend this expertise to the domain of agriculture for pest management and agroecological recommendations, and in the domain of ecology for understanding the interactions between landscape change and climate variability to support biodiversity conservation and ecosystem resilience. Through this interdisciplinary research direction, I hope to contribute towards scalable, evidence-based approaches for environmental sustainability and community-based natural resource management.

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